CLAIM-X

Healthcare Claim Fraud Detection & Investigation Risk Prediction

BDML PROJECT | AI23521 – Build and Deployment of Machine Learning Applications

PROJECT OVERVIEW

Claim-X is an AI-assisted review workbench for health-insurance fraud, waste and abuse teams. It combines rule-based checks with two machine-learning models, an Isolation Forest and a Gradient Boosting classifier, and a referral-network analysis. Together they turn raw claims into evidence-backed cases, rank those cases against the team's available hours, and give each case to a person to decide. All data is synthetic. The system only recommends; a human reviewer always makes the decision.

1. Problem Statement

Health insurers lose money to billing abuse such as duplicate billing, upcoding, services that were never delivered, and referral rings. Investigators receive far more alerts than they can review, and a bare score does not tell them why a claim was flagged or how to defend the finding to a provider or a regulator. Wrongly accusing a provider carries legal and reputational cost. Claim-X aims to point reviewers to the cases most worth their time, show the evidence behind every flag, and keep every decision explainable and recorded.

2. Project Objectives

- Detect suspicious claims and providers with rules, an unsupervised anomaly model, and referral-network analysis.
- Estimate an **investigation risk** (how likely a confirmed investigation is) for the next 30, 60 and 90 days.
- Group findings into cases with evidence IDs, rank them by priority against team hours, and route them to investigators.
- Present model outputs in a clear reviewer console with evidence, a network graph, a timeline and a written brief.
- Evaluate each model separately with appropriate classification and ranking metrics on held-out data.
- Expose everything through a REST API, secure it with role-based access, and support Docker deployment.

3. Machine-Learning Models

Model	Input and role	Expected output
Isolation Forest (unsupervised, scikit-learn)	Per-provider features (claims per day, average amount ratio, level-5 share, services per member per month, repeat-member ratio, referred-claim share, weekend ratio, volume trend). Features are scaled, and the forest scores how unusual each provider is against peers of the same specialty.	An anomaly score from 0 to 1 and a rank for every provider; the top 5% become findings, each with its three largest plain-language drivers.
Gradient Boosting classifier (supervised, scikit-learn)	The same provider features plus the count of prior confirmed investigations, taken at monthly cut-offs using only data up to each cut-off. One model is trained for each window (30, 60, 90 days) on labelled outcomes.	An **investigation risk** between 0 and 1 for each window, a band (Low below 0.3, Medium below 0.6, High from 0.6) and the top three drivers.

The two models have separate roles: the Isolation Forest needs no labels and ranks providers by how unusual they look, while the Gradient Boosting classifier learns from labelled investigation outcomes. Gradient Boosting settings: 100 trees, learning rate 0.1, maximum depth 2, subsample 0.8, seed 42. Isolation Forest settings: 200 trees, contamination 0.05, seed 42.

The system also contains **seven rule-based detectors** (duplicate billing, unbundling, upcoding, billing during a hospital stay, excess utilization, impossible timing, repeat investigation history) and a **referral-network analysis** (NetworkX graph with Louvain communities). These are deterministic and are not presented as machine-learning predictions.

> The risk band comes from the model. In one case a stated policy raises it: a provider with a prior confirmed investigation and rising volume is placed in the High band, and the screen says so (*band_source: escalated*). The model's own number is never changed. This follows the rule that fixed rules must not be presented as machine-learning output.

4. Proposed Workflow

1. **Data** – a synthetic claims history is generated with a fixed seed (5,000 claims, 40 providers, 15 facilities, 300 members, 600 referrals, six months of data) with planted fraud scenarios and honest decoys.
2. **Detection** – the seven rules, the Isolation Forest and the referral-network analysis each raise findings, and every finding carries claim IDs and a plain-language reason.
3. **Risk prediction** – Gradient Boosting estimates the 30-, 60- and 90-day investigation risk from the same feature definitions used in training.
4. **Cases and ranking** – findings are grouped into cases and ranked by risk, dollars, members affected, severity and evidence strength against the team's hours.
5. **Brief** – a written brief cites every statement to its evidence; an LLM may help write it, but only from masked data, and a template is the fallback.
6. **Review** – a team lead assigns the case, the investigator decides with a written reason, and every step is recorded in an append-only audit log.
7. **Results display** – the console shows the evidence, graph, timeline, the three risk windows and the recommended action, each clearly separate.

5. Technology Stack

Layer	Used
Machine learning	scikit-learn (Isolation Forest, Gradient Boosting), NetworkX and python-louvain (referral graph)
Data processing	Python 3.11, pandas, NumPy, SQLite
Backend / API	FastAPI REST API, Pydantic, JWT sign-in (HS256) with bcrypt passwords, role-based access
AI-written briefs	Groq (tested live), Claude and Grok (mocked tests), with masking, validation and caching
Frontend	React 19, Vite, Tailwind CSS, Recharts, react-force-graph; light and dark themes
Reports and alerts	reportlab (PDF and CSV reports), AWS SNS (urgent-case email)
Quality	pytest (527 tests, 97% coverage, no network access), Vitest (133 tests), ruff, a real-Chrome walkthrough, GitHub Actions CI
Deployment	Docker image serving the API and the web app on one address; Render blueprint

6. Model Evaluation

Each model was evaluated separately. The Gradient Boosting models were trained on earlier monthly cut-offs and tested on the last cut-off whose prediction window was fully observed, so the test data was never used in training. The risk band threshold of 0.3 (the Medium band) is used to turn a risk into a yes/no decision for the classification metrics.

6.1 Gradient Boosting (investigation risk), held-out test

Window	Train rows / positives	Test rows / positives	Accuracy	Precision	Recall	F1	Macro-F1	ROC-AUC	PR-AUC (base rate)
30 days	120 / 12	40 / 3	0.850	0.000	0.000	0.000	0.459	0.685	0.145 (0.075)
60 days	120 / 18	40 / 8	0.825	0.600	0.375	0.462	0.679	0.785	0.642 (0.2)
90 days	80 / 15	40 / 10	0.775	0.545	0.600	0.571	0.709	0.767	0.627 (0.25)

30-day	Predicted: no	Predicted: yes
Actual: no	34	3
Actual: yes	3	0

60-day	Predicted: no	Predicted: yes
Actual: no	30	2
Actual: yes	5	3

90-day	Predicted: no	Predicted: yes
Actual: no	25	5
Actual: yes	4	6

Confusion matrices at threshold 0.3 (rows are actual, columns are predicted).

> **How to read these results.** Each test set has only 40 providers and between 3 and 10 positives, so one provider moves the numbers a great deal. Plain accuracy is misleading here: a model that always answers "no" would score 92.5% at 30 days, 80% at 60 days and 75% at 90 days. ROC-AUC (0.69, 0.79, 0.77) and macro-F1 give a fairer picture. The 30-day model barely beats chance, and its recall of 0 means it missed all three positives at the 0.3 threshold. The longer windows score higher partly because more investigations fall inside a longer window (higher base rate). These results validate the pipeline on synthetic data. They are not a claim of real-world accuracy.

6.2 Isolation Forest (provider anomaly), ranking against planted providers

The Isolation Forest is unsupervised, so it is evaluated by how it ranks the providers that were deliberately planted, out of 40 providers.

Planted provider	ID	Rank by anomaly score
Referral-ring referrer	PRV-A01	1
Over-utilizing physiotherapy provider	PRV-019	2
Upcoding provider	PRV-005	3
Provider billing 30 hours in one day	PRV-029	4
Repeat offender with rising volume	PRV-010	5
Honest busy specialist (decoy, must not rank high)	PRV-015	8

All five planted providers occupy the top five ranks (precision at 5 = 1.0, recall at 5 = 1.0), and the honest busy specialist ranks 8th, outside the top five. Only providers above the 95th percentile (2 providers) become findings.

6.3 Rule-based detectors and referral analysis, planted-scenario coverage

Planted scenario	Planted	Flagged	Caught by
Duplicate billing (claims)	20	20	duplicate rule
Unbundled panels (claims)	50	50	unbundling rule
Phantom services (claims)	11	11	billing-during-stay rule
Over-utilizing members	3	3	excess-utilization rule
Upcoding provider	1	1	upcoding rule
Repeat offender	1	1	repeat-history rule (30-day band High)
Impossible timing (30 hours in a day)	1	1	impossible-timing rule
Referral ring (5 entities)	5	5	referral-network analysis and anomaly model
Honest busy specialist and 40 honest same-day follow-ups	41	0	none (correct)

7. Deployment and API

Item	Detail
REST API	FastAPI; endpoints include */health*, */auth/login*, */overview*, */queue*, */cases/{id}* (evidence, graph, brief, three-window risk), */cases/{id}/decision*, */cases/{id}/assign*, */cases/{id}/report.pdf*, */audit* and admin endpoints; interactive documentation at */docs*
Access control	Admin, team lead and investigator roles are enforced by the API; every refusal returns 403 and is audited
Container	A Dockerfile builds the web app and serves it with the API on one address; settings come from environment variables and no secret is stored in the image
Hosting	A Render blueprint (*render.yaml*) deploys the container; a CI job builds the image and smoke-tests it
Testing	527 backend tests run with no network access (97% coverage); 133 frontend tests; an end-to-end walkthrough signs in as each role in Chrome

8. Expected Outcome

A working prototype that combines rules, an anomaly model, network analysis and investigation-risk prediction in one reviewer console, with every flag tied to evidence, every decision tied to a person and a reason, and every step audited. It is a decision-support aid for investigators, not a determination that any provider committed wrongdoing, and it is demonstrated on synthetic data.

9. Screenshots of the Application

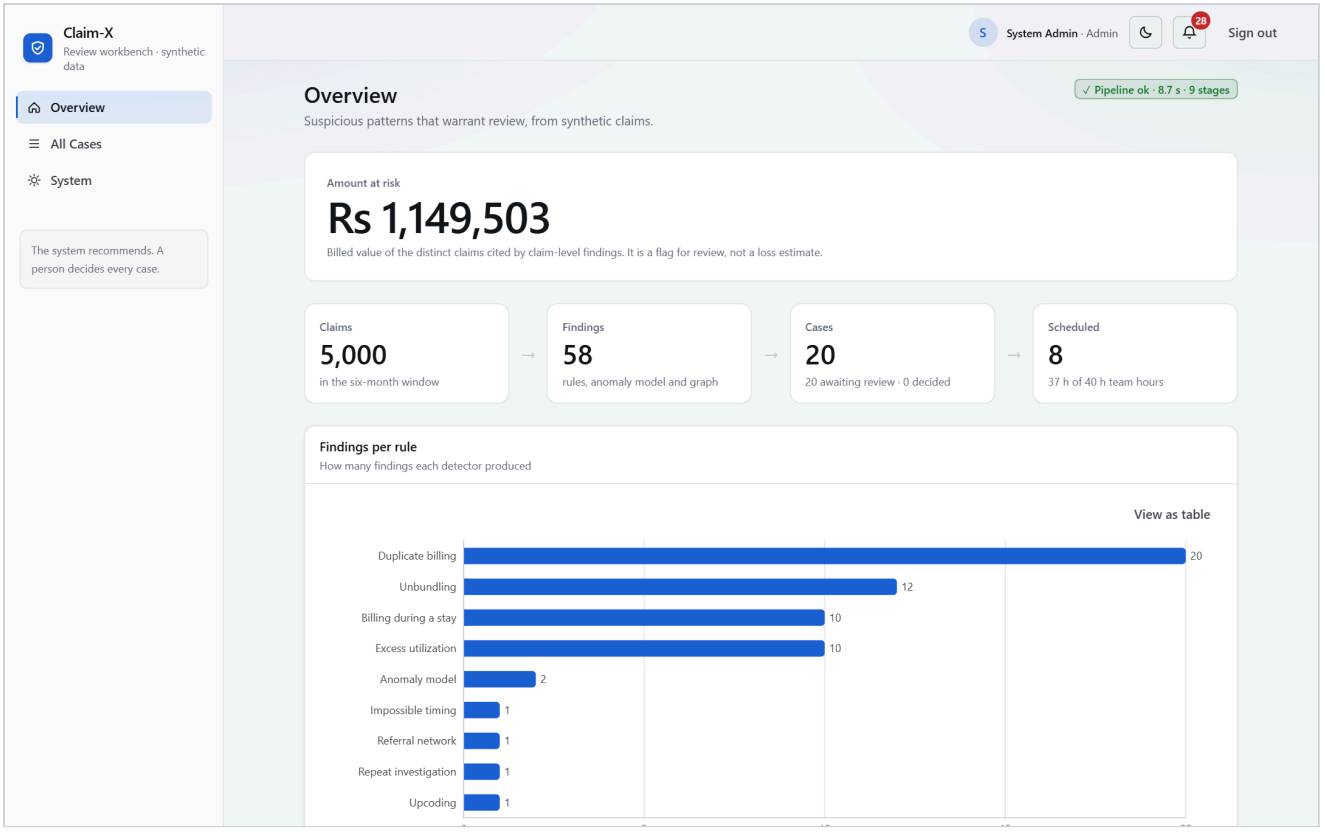


Figure 1. Overview: claims scanned, findings, cases, amount at risk, and findings per rule.

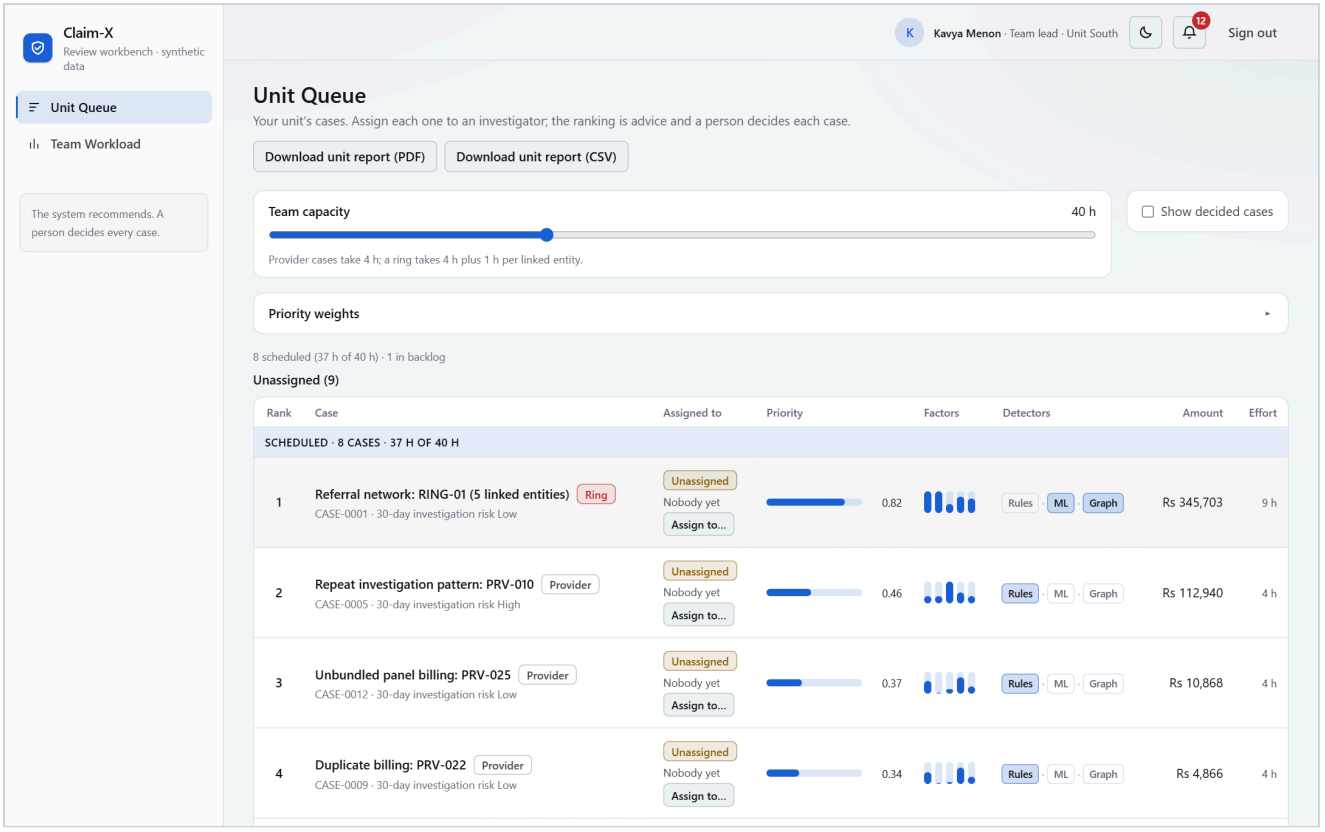


Figure 2. Unit queue for a team lead: cases ranked by priority and scheduled against team hours.

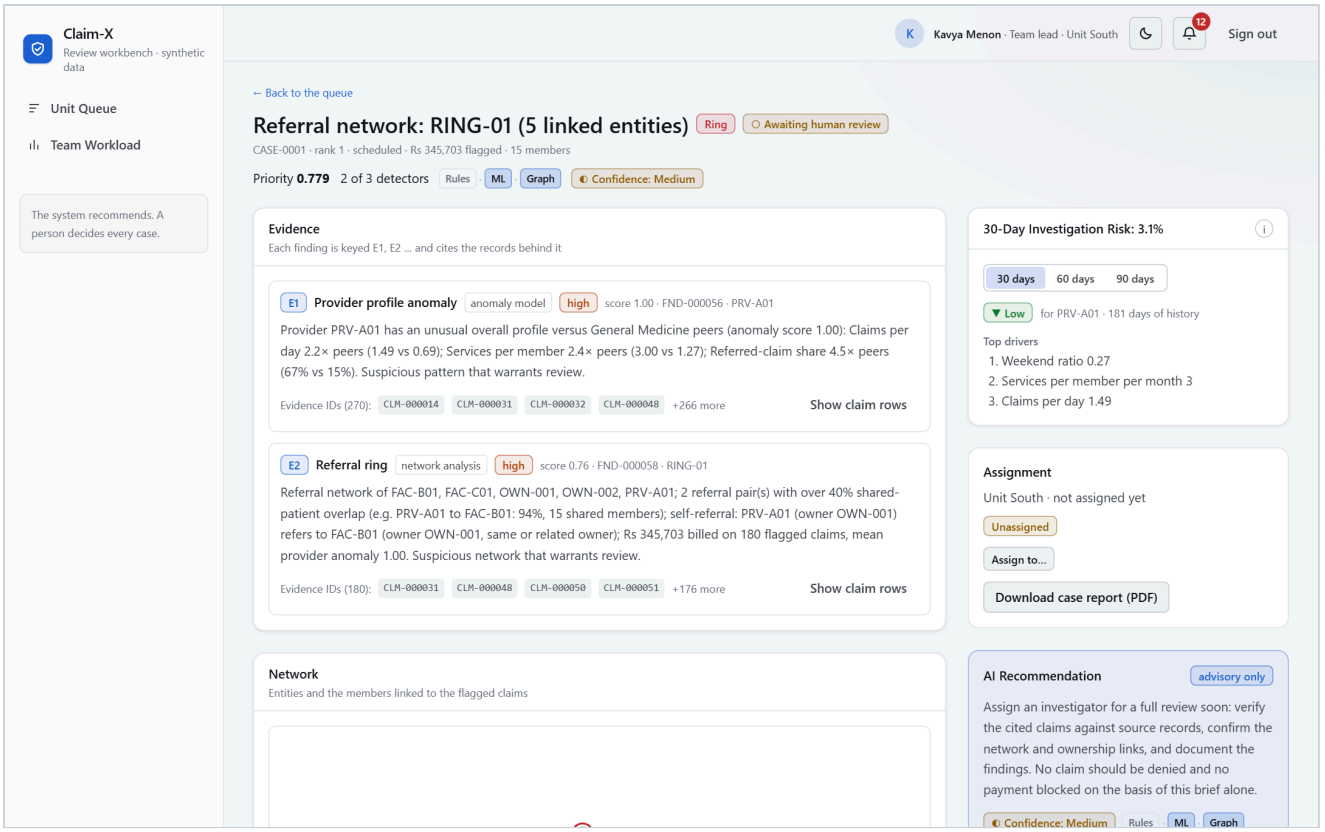


Figure 3. Case detail: evidence items that name the type of check, detector chips, risk panel and assignment.

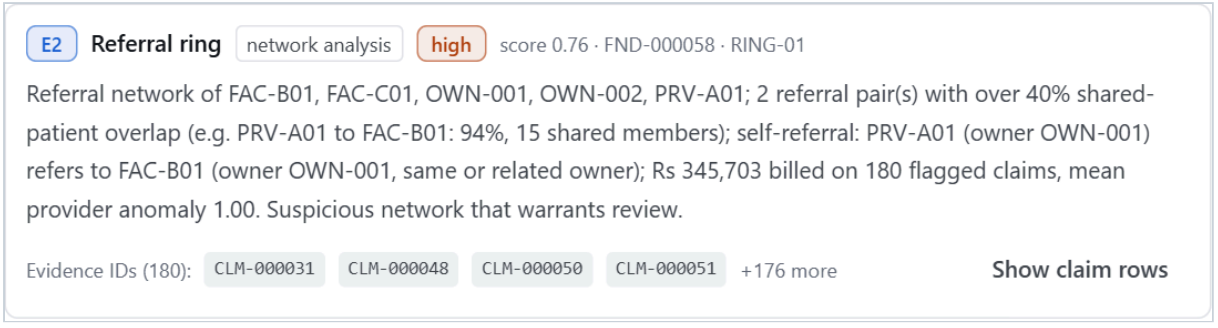


Figure 4. One evidence item (referral ring), with its reason and the claims behind it.

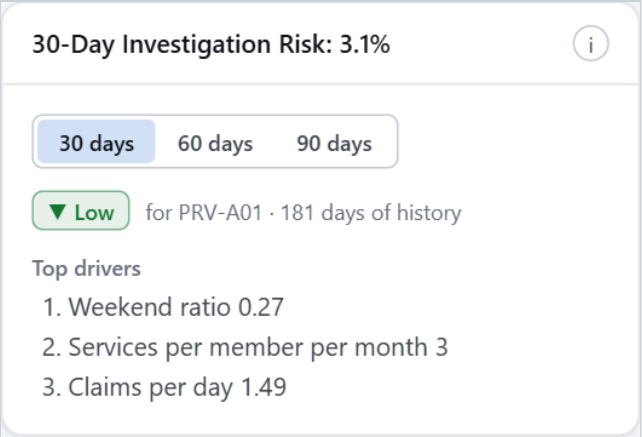


Figure 5a. 30-day investigation risk, band and drivers.

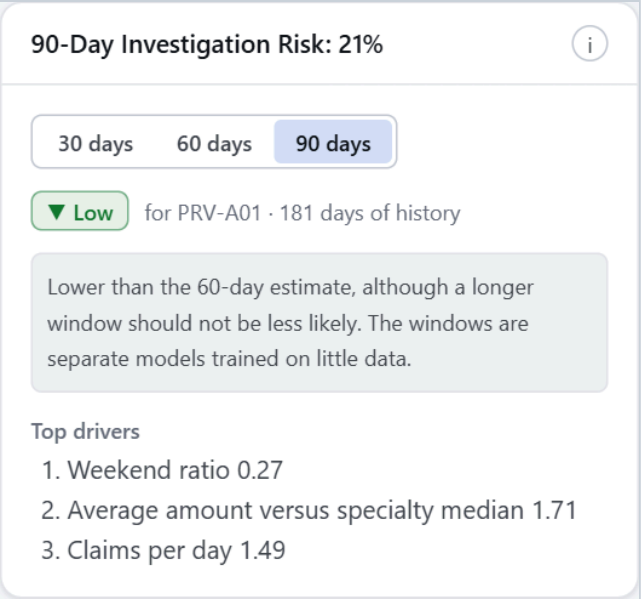


Figure 5b. The same panel switched to the 90-day window.

Brief

Written from the evidence only; citations open the evidence above

Source **Template** Written by the built-in template: LLM_PROVIDER is none.

Investigation brief: CASE-0001

Case type: ring (RING-01). Priority rank 1, queue scheduled. Status: Awaiting human review.

Summary

Case CASE-0001 concerns 5 linked entities (FAC-B01, FAC-C01, OWN-001, OWN-002, PRV-A01).

2 of 3 detector groups fired: anomaly (provider profile anomaly) [E1]; graph (referral ring) [E2].

The flagged claims total Rs 345,703 and involve 15 members [E2].

The behaviour is suspicious and warrants review by a human investigator; it is not a finding of wrongdoing.

Evidence

- [E1] Provider profile anomaly (anomaly model), high, score 1.00: Provider PRV-A01 has an unusual overall profile versus General Medicine peers (anomaly score 1.00): Claims per day 2.2× peers (1.49 vs 0.69); Services per member 2.4× peers (3.00 vs 1.27); Referred-claim share 4.5× peers (67% vs 15%). Suspicious pattern that warrants review. Scope: provider-level signal across 270 claims of PRV-A01.
- [E2] Referral ring (network analysis), high, score 0.76: Referral network of FAC-B01, FAC-C01, OWN-001, OWN-002, PRV-A01; 2 referral pair(s) with over 40% shared-patient overlap (e.g. PRV-A01 to FAC-B01: 94%, 15 shared members); self-referral: PRV-A01 (owner OWN-001) refers to FAC-B01 (owner OWN-001, same or related owner); Rs 345,703 billed on 180 flagged claims, mean provider anomaly 1.00. Suspicious network that warrants review. Claims: 180 (for example CLM-000031, CLM-000048, CLM-000050).

Timeline

- 2026-01-02 to 2026-01-26: 30 ring claims worth Rs 58,435 in 2026-01 [E2]
- 2026-02-02 to 2026-02-23: 30 ring claims worth Rs 56,922 in 2026-02 [E2]
- 2026-03-02 to 2026-03-29: 30 ring claims worth Rs 58,574 in 2026-03 [E2]
- 2026-04-03 to 2026-04-22: 30 ring claims worth Rs 58,268 in 2026-04 [E2]
- 2026-05-04 to 2026-05-29: 30 ring claims worth Rs 58,593 in 2026-05 [E2]
- 2026-06-02 to 2026-06-27: 30 ring claims worth Rs 54,911 in 2026-06 [E2]

Figure 6. The evidence-cited brief, with its source badge.